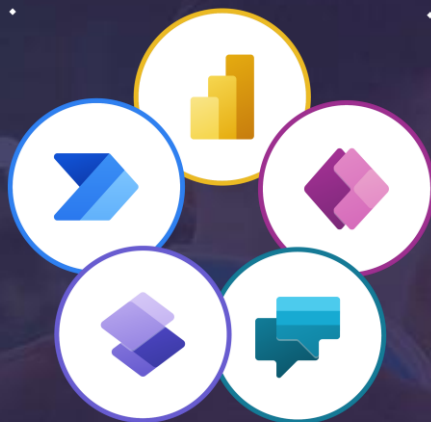


5th



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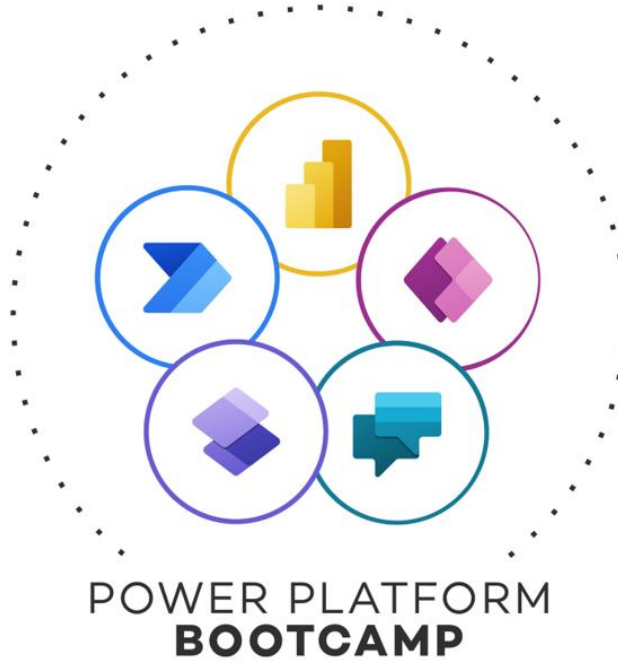
Data Modelling 101 for PowerBI newbies



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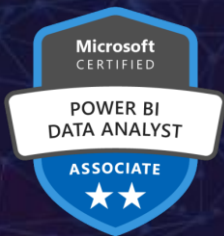
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- BI Consultant at Seidor Analytics
- PL-300 & DP-900 certified



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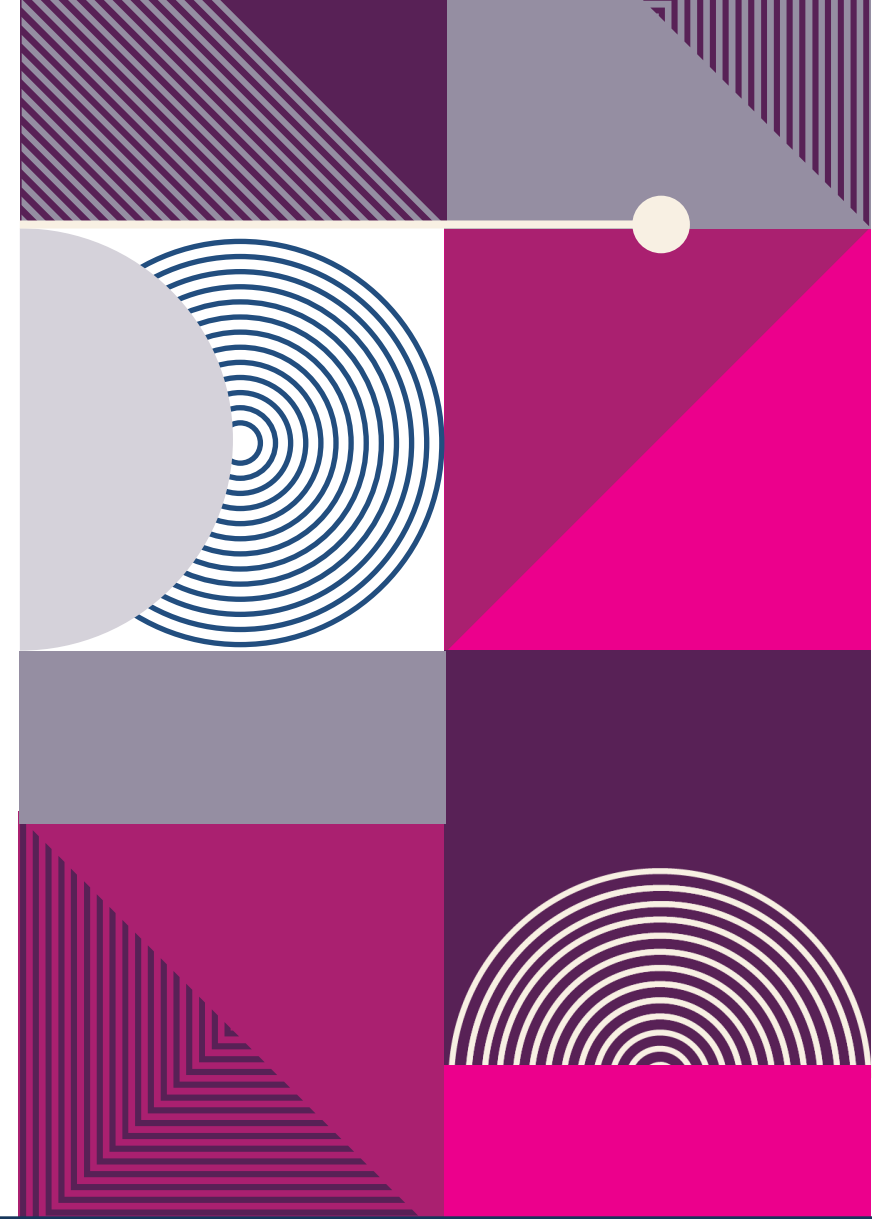
DATA MODELLING 101 FOR POWERBI NEWBIES



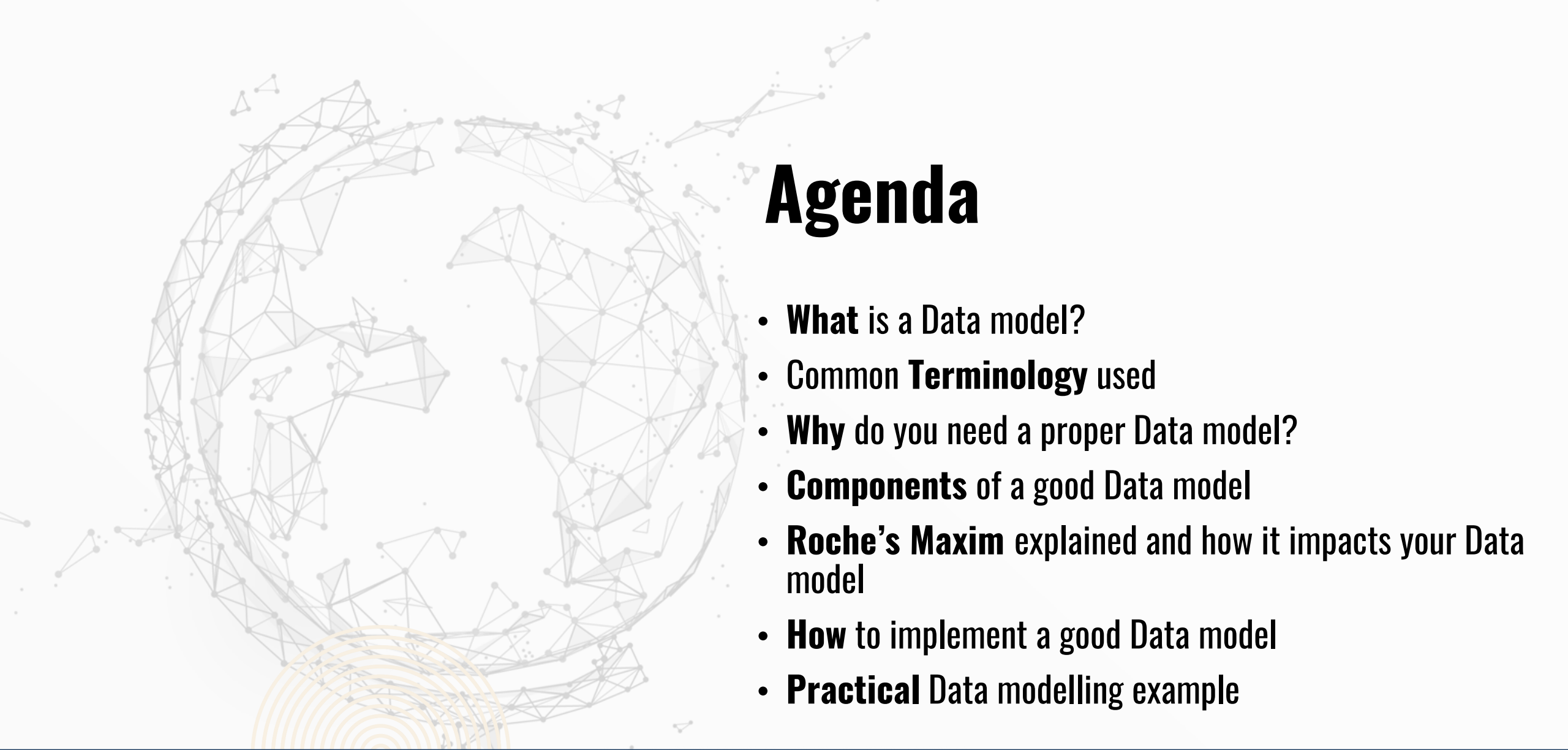
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EXPECTATIONS & LIMITATIONS

- The user has **introductory experience** in PowerBI
- The user can **import data** into PowerBI
- Due to the complexity and limited time available during the session, this Data modelling session will only be of an **introductory** nature.
- Data modelling will change from Project to Project-engage with a qualified BI Consultant within your organisation to assist you when you are stuck.
- The views expressed in this session are my own



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Agenda

- **What** is a Data model?
- Common **Terminology** used
- **Why** do you need a proper Data model?
- **Components** of a good Data model
- **Roche's Maxim** explained and how it impacts your Data model
- **How** to implement a good Data model
- **Practical** Data modelling example

WHAT IS A DATA MODEL

The “official” Microsoft definition of a Data model is:

“Data modelling is the process of analysing and defining all the different data types your business collects and produces, as well as the relationships between those bits of data.”

<http://tinyurl.com/data-model-def>



Microsoft



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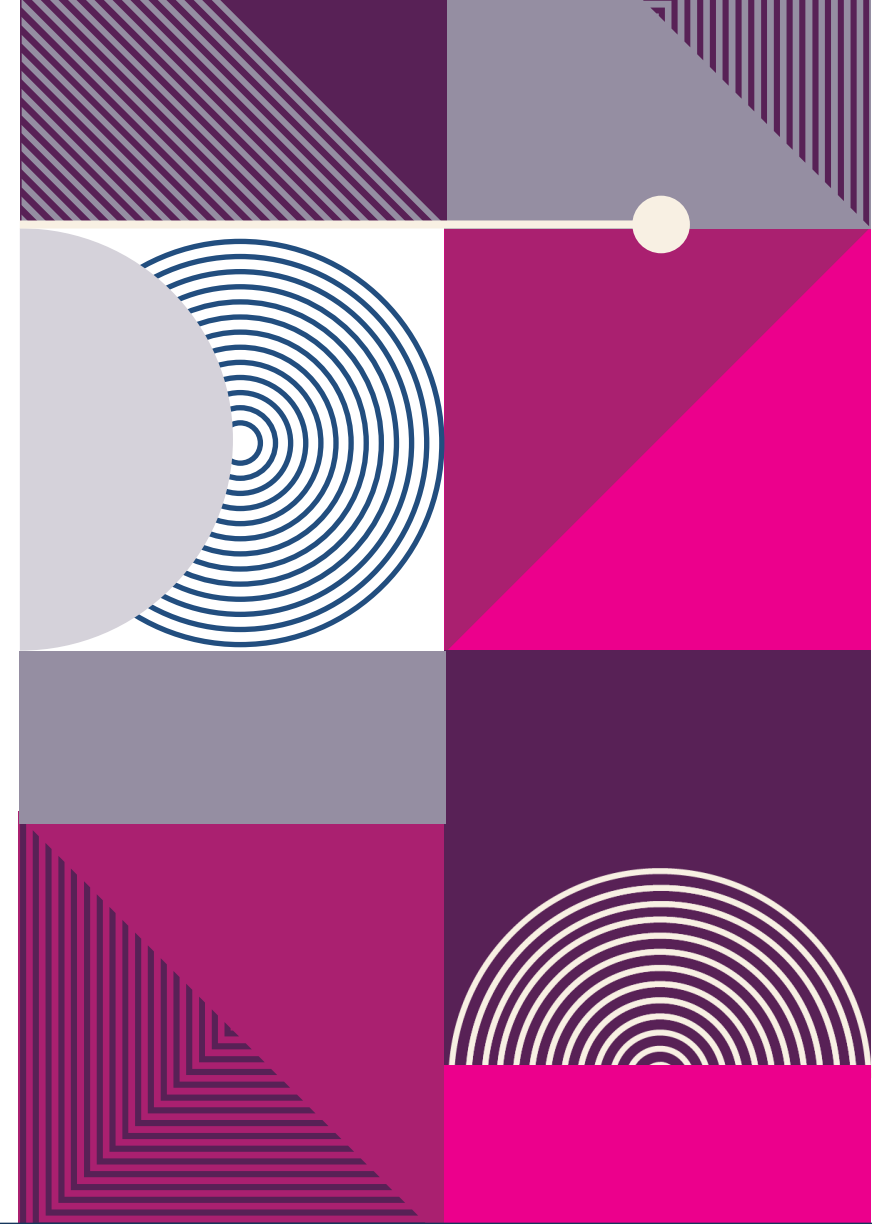
WHAT IS A DATA MODEL

In “*Layman’s terms*”, a Data model within PowerBI will consists of the below elements

- Imported **data** from numerous sources i.e.:Tables
- **Relationships** between these imported tables
 - Differentiation between Rows and Columns
- **Measures** created
- **Calculated columns** created

The Data model complexity can differ based on import mode and the cardinality of imported data

Visual representation of how data is organized and structured to support business objectives



TERMINOLOGY

DIMENSION TABLE

Store descriptive attributes or dimensions that provide context to your data (in Fact Table)

FACT TABLE

“Repo” for quantitative measurable data (Numerical data to be analyzed)

RELATIONSHIPS

Relationships formed between the Dimension & Fact tables

PRIMARY KEY

Unique identifier within Dimension table

FOREIGN KEY

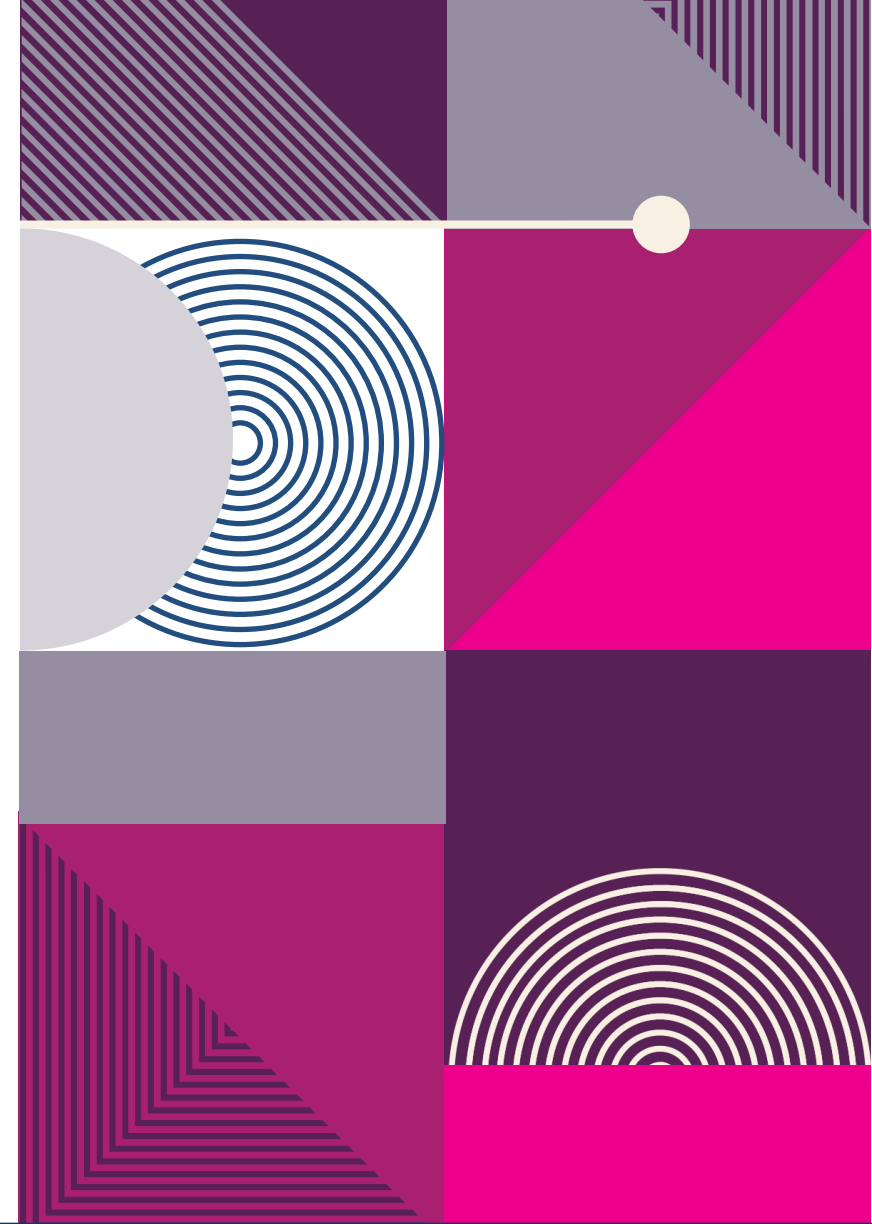
Contain foreign keys that references Dimension tables

MEASURES

Dynamic calculations created within model

WHY DO YOU NEED A DATA MODEL?

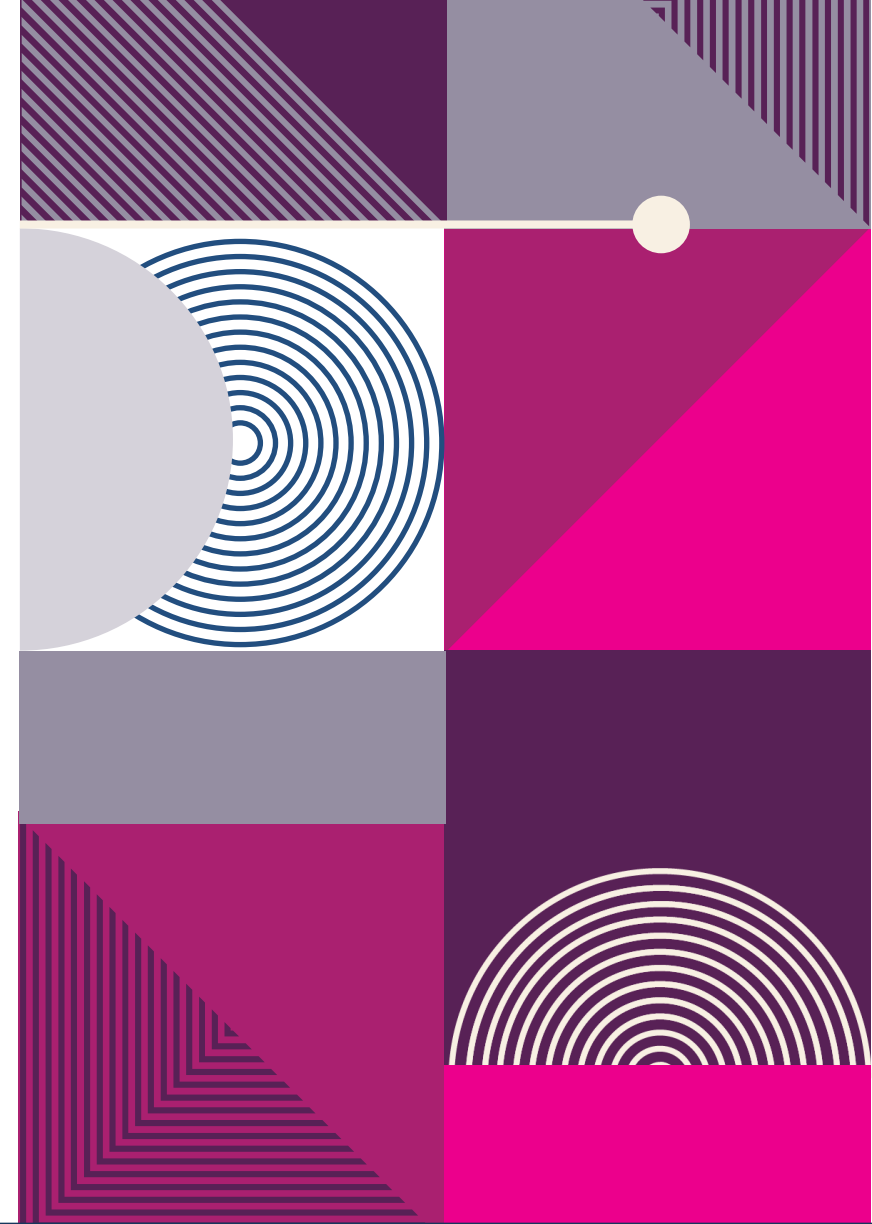
- Organization
- Performance
- Data integrity
- Improved Analysis
- Lay foundation for advanced analytics
- Better and faster performing DAX (and DAX syntax)



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VALUE VS HASH & RUN LENGTH ENCODING

- Ever wondered how PowerBI can perform calculations so quick?
- PowerBI use various types of encoding which basically “compresses” the data, to ensure optimal performance
- This is why confirming and assigning the correct Data Type is so important
 - Saving numerical data would use the “least” space
- **Resources to learn more about it on next page**



ENCODING ADDITIONAL RESOURCES



**YOUTUBE
POWERBI TIPS**

[https://tinyurl.com/
PowerBI-tips-Encoding](https://tinyurl.com/PowerBI-tips-Encoding)

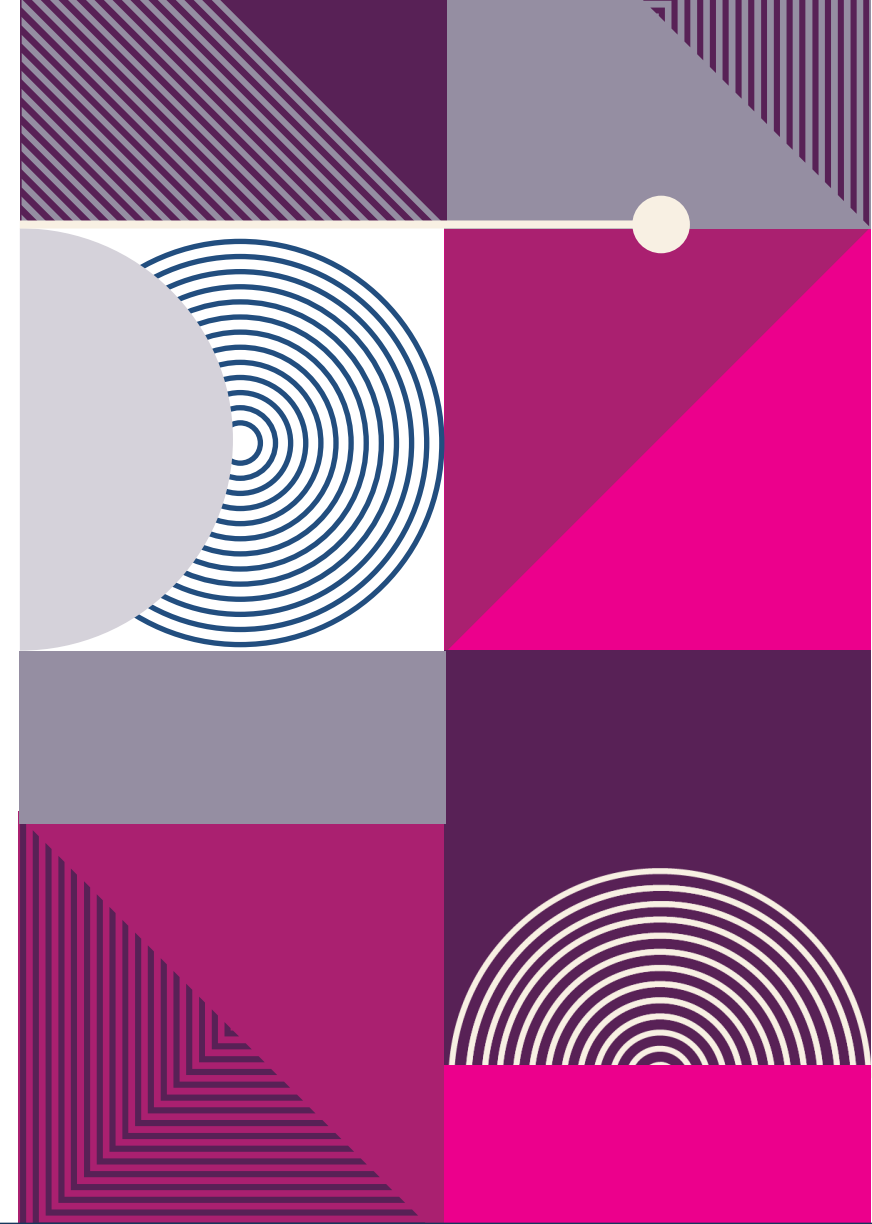


ENTERPRISE DNA

[https://tinyurl.com/
EDNA-Encoding](https://tinyurl.com/EDNA-Encoding)

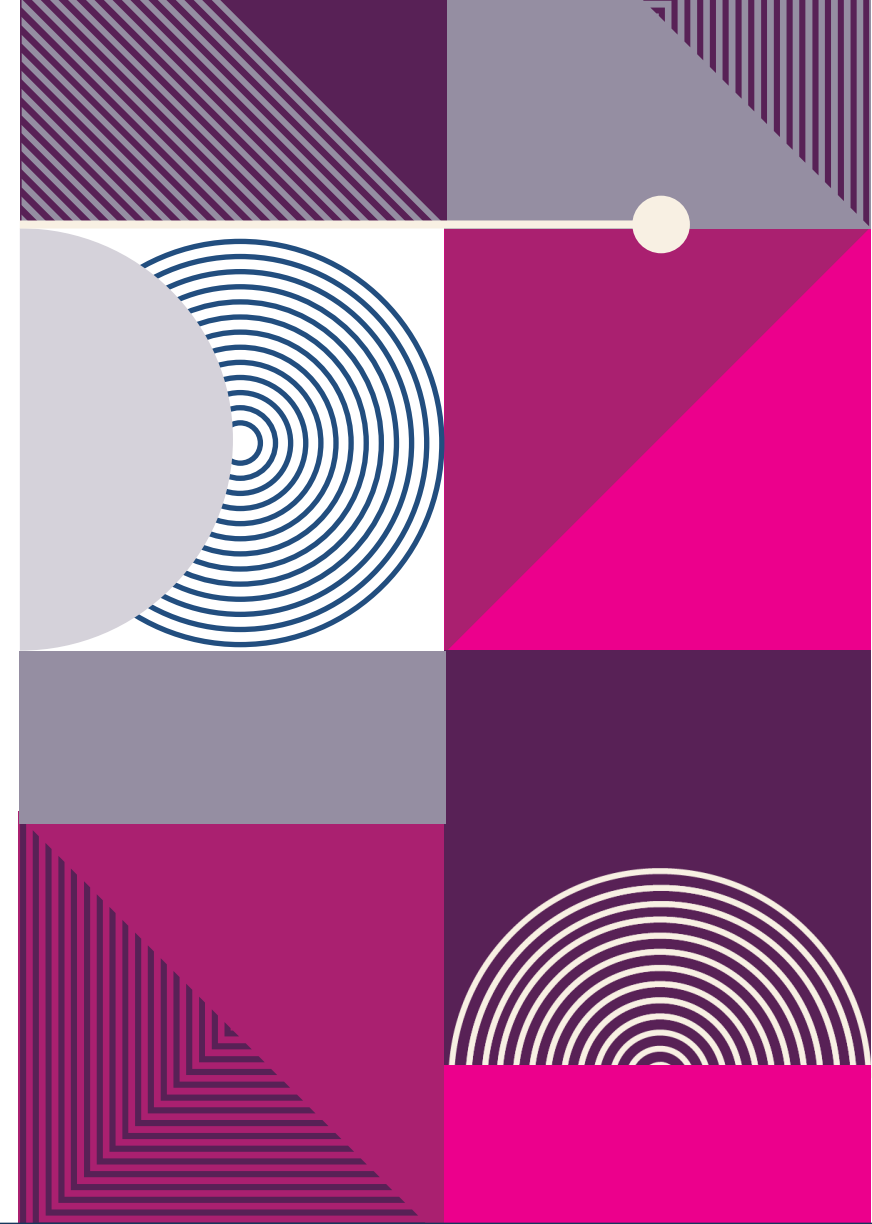
COMPONENTS OF A GOOD DATA MODEL?

- **Data** to be analyzed will depend on the **import mode**. Numerous considerations must be considered and made.
 - Import, Direct Query or Hybrid mode
 - Refresh rate: Real time analytics vs Daily/ Weekly etc
 - Possible Data aggregations for Historical Data
- PowerBI service file limitations



COMPONENTS OF A GOOD DATA MODEL?

- **Dedicated Date Table** for Time intelligence calculations
- **Relationships**
 - One to many
 - Limit many to many relationships (specific use cases)
 - Limit one to one relationships (specific use cases)
 - Ensure directions are set properly
 - Dimensions **always** filter the Facts
- Star or Snowflake schema arrangement
 - Traditional layout or Collie/ Waterfall layout



COMPONENTS OF A GOOD DATA MODEL?

- **Measures**
 - No additional space used within the Data model
 - Only evaluated once used
 - Best practice- keep measures in Measure folders
- **Calculated columns**
 - Takes up additional space within the Data model
 - Calculates each time the Data model refreshes
 - Try to do Calculated columns/ Conditional columns in PowerQuery

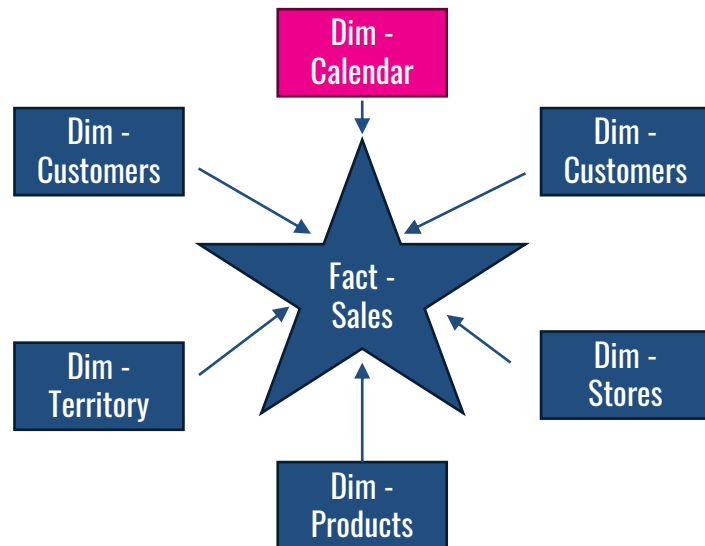


Microsoft

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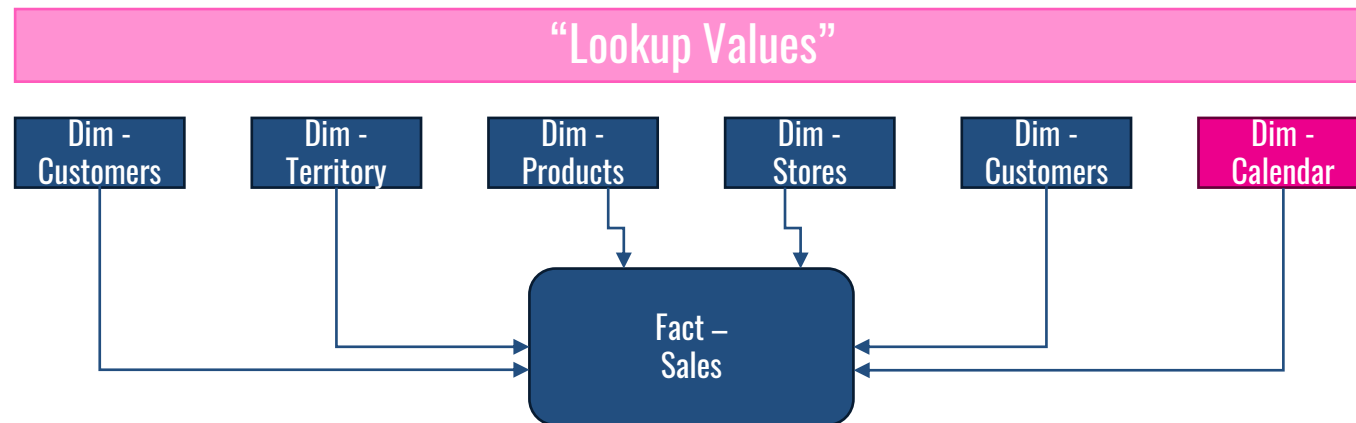
COMPONENTS OF A GOOD DATA MODEL?

Fact Table placed in the centre with Dimension tables placed around the Fact table



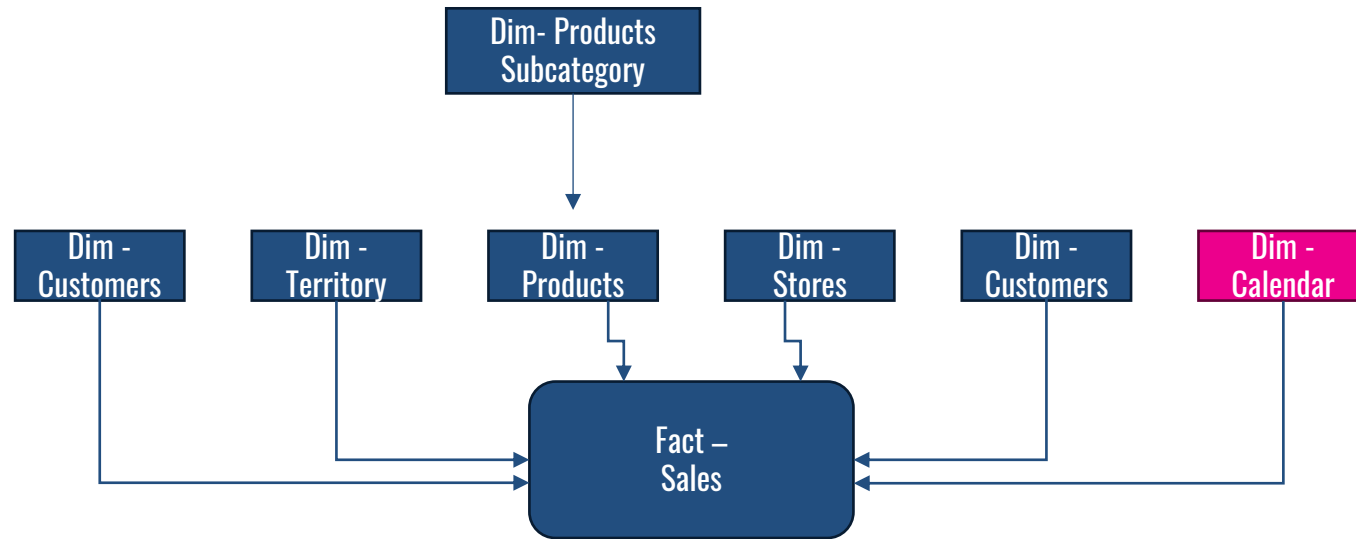
COMPONENTS OF A GOOD DATA MODEL?

Collie or Waterfall layout- this remains a Star Schema-ONLY the layout changes
Fact table(s) placed at the bottom , and Dimension table(s) placed at the top.
This is a visually effective way to illustrate how the Data filters (downstream) to the Fact table.



COMPONENTS OF A GOOD DATA MODEL?

Fact Table placed in the centre with Dimension tables placed around the Fact table- additional “dimension tables”- further normalization happens.



COMPONENTS OF A GOOD DATE TABLE

MARK Date Table

Cover ALL Dates in Fact(s)

Date Column span FULL years

UNIQUE values

NO blanks in range

NO Missing Dates

Date Data type column **MUST** be included



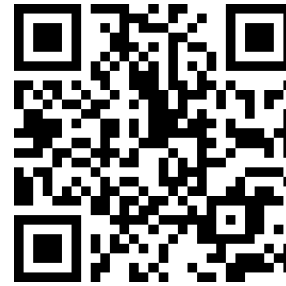
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CUSTOM DATE TABLE RESOURCES



ENTERPRISE DNA

[https://tinyurl.com/
Custom-Date-Table-EDNA2](https://tinyurl.com/Custom-Date-Table-EDNA2)



GORILLA BI

[http://tinyurl.com/
Custom-Date-Table-BI-Gorilla](http://tinyurl.com/Custom-Date-Table-BI-Gorilla)

**CALENDAR
CALENDARAUTO
GENERATESERIES**

DAX

Numerous DAX
Statements can be used

COMPONENTS OF A GOOD DATA MODEL?

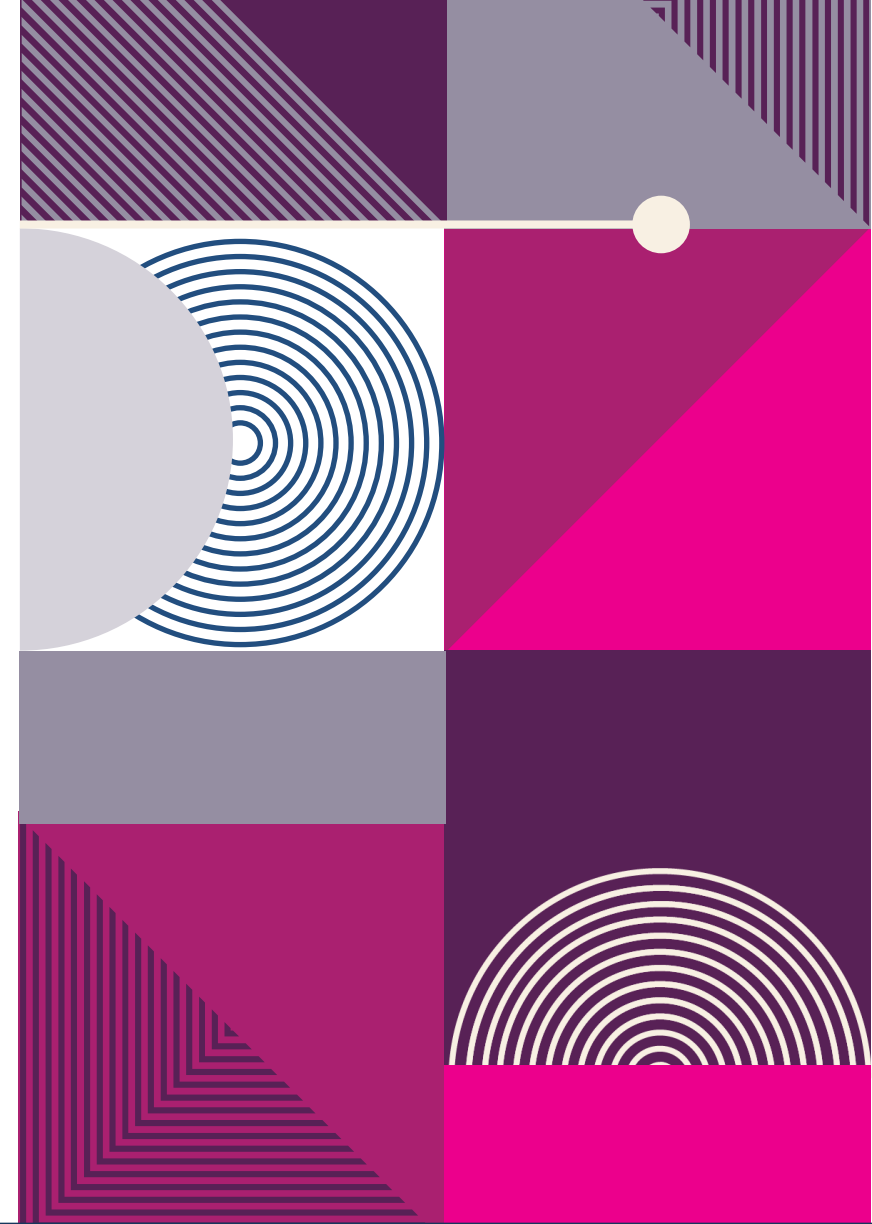
Dimension tables

- Contains the descriptive attributes/ information that could filter (slice & dice) or provide more details about the Fact tables
- Wide – many columns, but fewer rows
- Contains the Primary Keys

COMPONENTS OF A GOOD DATA MODEL?

Fact table

- Contains dimensions for the attributes
- Many rows and few columns
- Usually contains the numeric data
- Contains the Foreign keys



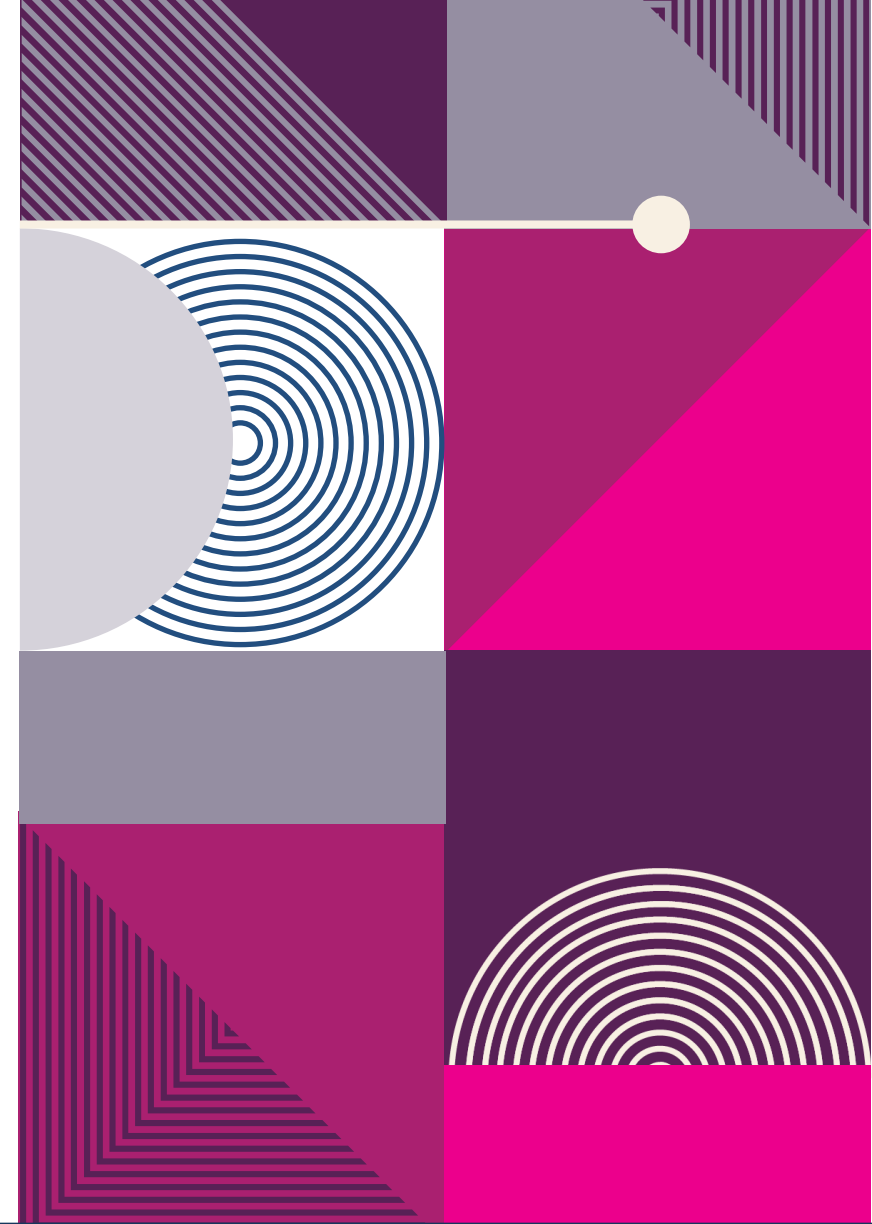
COMPONENTS OF A GOOD DATA MODEL?

Relationships direction

- One to Many
- One to One
- Many to Many

Relationship status

- Active
- Inactive (use Userelationship)



COMPONENTS OF A GOOD DATA MODEL?

Measures

- No cell context, use column context
- Measure branching of measures
- Naming conventions
- Implicit vs Explicit
- Use of variables
- Measure tables



Microsoft

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ROCHE'S MAXIM EXPLAINED

Roches Maxim state that “Data should be transformed as far upstream as possible and as far downstream as necessary.

- Upstream = where data is produced
- Downstream = where data is consumed

Possible query folding as well

Source

Power Query

DAX

Calculated Columns



Microsoft

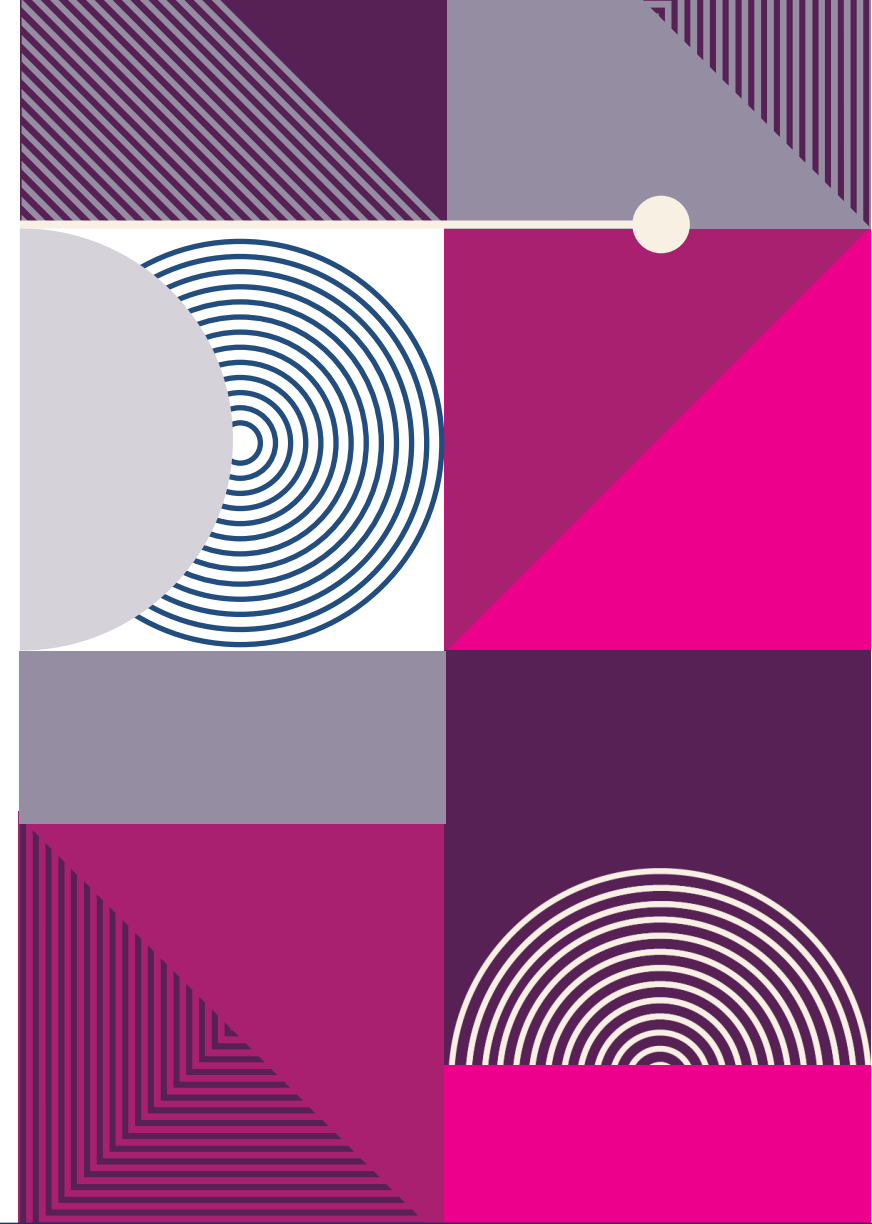
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HOW TO IMPLEMENT A GOOD DATA MODEL

Data import – correct mode selected + PowerBI service limitations

ETL vs ELT considerations

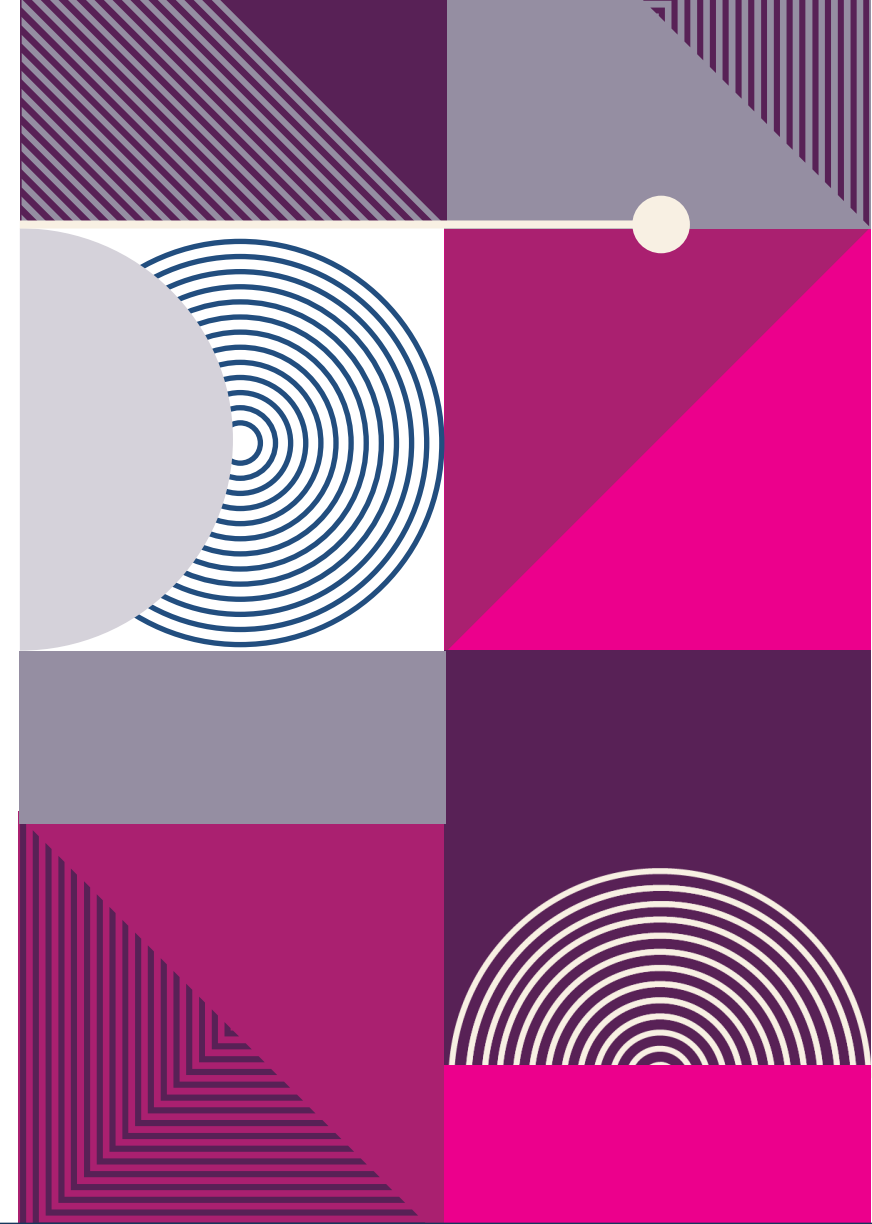
Always follow recommended best practices/ internal company standard on data modelling i.e: Import mode, normalization/ denormalization, naming conventions



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PRACTICAL DATA MODELLING EXAMPLE

We'll use a simple Dataset to showcase some simple Data modelling concepts.



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FURTHER DATA MODELLING LEARNING



**MICROSOFT
LEARN**

[http://tinyurl.com/
MS-Learn-Data-Modelling](http://tinyurl.com/MS-Learn-Data-Modelling)

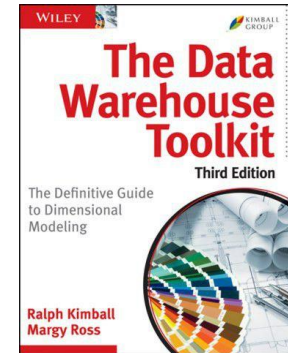
Official Microsoft Learn and
learning paths



SQLBI

[http://tinyurl.com/
SQLBI-Free-Course](http://tinyurl.com/SQLBI-Free-Course)

Free Data modelling
course by SQLBI on
basic Data modelling



KIMBALL

Probably the best resource to
learn more about Relational
Data modelling in general

BONUS SECTION (RECAP)



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DATA MODELLING “SINS”

Duplication of Data

Ambiguous relationships

Too many tables

Absence of Documentation

Missing measures

Overcomplicated Calculations

Disregard for Data integrity

Excessive denormalization

Lack of Load optimization



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Any Questions?



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